Database Migration to Huawei Cloud



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Why Migrate Databases to the Cloud?

Cost savings, scalability, and managed services

Practical guidance, not a formal delivery commitment

Key Driver Eliminate expensive Oracle/SQL Server licensing — 60-80% TCO reduction with GaussDB

1. Cost Savings

60-80% TCO reduction

- Oracle Enterprise = $47K/core + $10K/year support
- GaussDB = $0.14/hour — pay-as-you-go
- No licensing audit overhead

2. Scalability

Horizontal scale, distributed DBs

- GaussDB, GeminiDB for distributed scale
- DWS columnar engine for analytics
- Auto-scaling without downtime

3. Managed Services

No DBA overhead

- Automated patching, backup, HA
- DAS dashboard for monitoring
- 99.95% SLA guaranteed

4. Data Sovereignty

In-country LATAM regions

- Meet regulatory requirements
- Data stays in-country
- Compliance-ready architecture

Cloud-native databases deliver immediate cost savings and long-term strategic advantages for LATAM enterprises.

Track A: Transactional Database Migration

5-Step Methodology

Step 1 — Schema (UGO, 1-2d)
Step 2 — Data (DRS, 4-8h/100GB)
Step 3 — App Adapt (AI, 1-2 sprints)
Step 4 — Testing (1 week)
Step 5 — Cutover (<30 min)



Target Databases

GaussDB (MySQL/PG compatible)
RDS for MySQL/PostgreSQL
Babelfish for TDS compatibility
DRS full + incremental sync



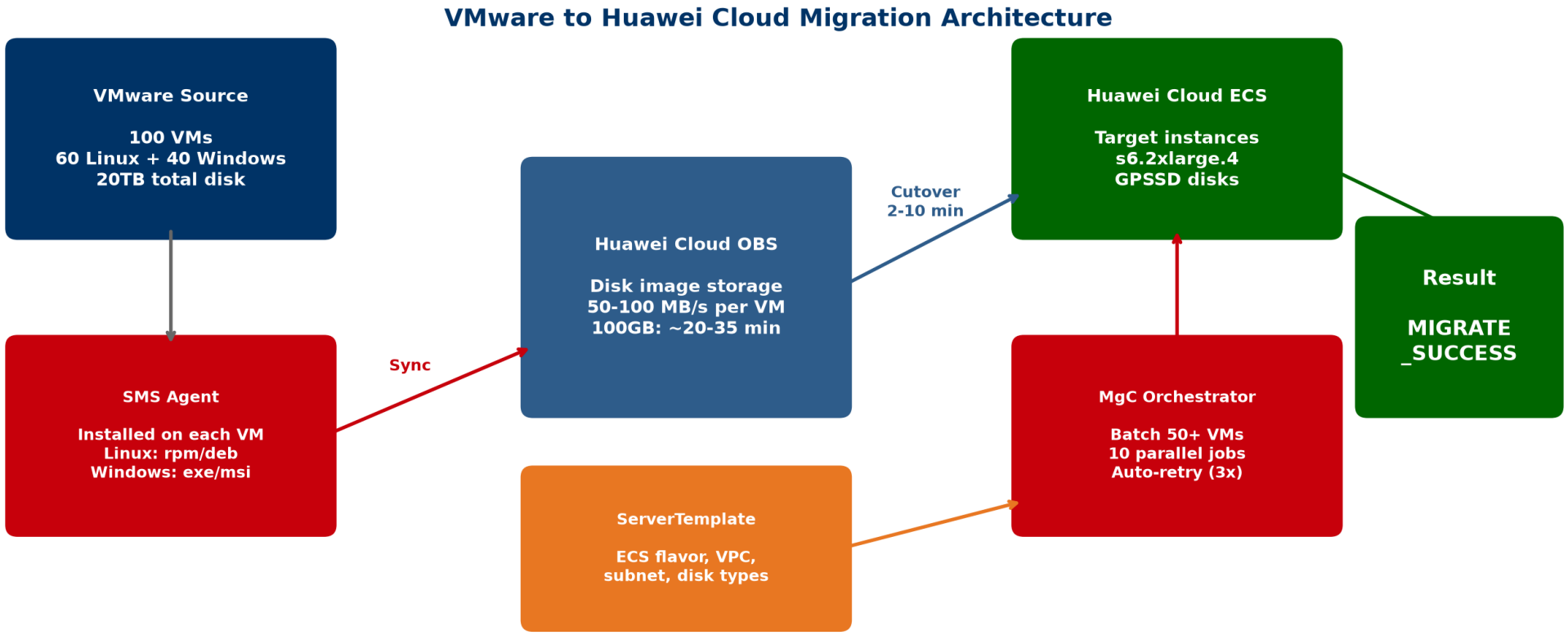
Key Metrics

80% schema auto-conversion
70% SQL rewrite via AI
99.99% data consistency
<30 min cutover window



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Track A Demo: Oracle & SQL Server Migration



Banking sector migration: Oracle 19c → GaussDB via UGO+DRS. 1,200 PL/SQL objects converted, 15,000 lines translated. DRS: 120GB/hour, 4 parallel threads. Cutover: <30 min with incremental sync.

Source: UGO Schema Conversion + DRS Data Replication Service

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Track B: Data Warehouse Migration

What to Ask AI

Example: "Convert our Oracle Exadata star schema to DWS. Generate ELT pipelines using DataArts Studio. Optimize distribution keys for our top 20 queries."



AI Should Know & Verify

DWS MPP columnar architecture, distribution key strategy, OBS foreign table loading. Result: 3-5x query performance vs. on-prem Exadata, 70% cost reduction.



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Big Data Platform Migration to MRS

Migration Job Execution: Two-Phase Process

Phase 1: Background Sync

- Agent uploads disk data to OBS
- Non-disruptive — VM stays running
- Sync speed: 50-100 MB/s per VM
- 100GB disk: ~20-35 min
- Monitor: sync %, data transferred, ETA

Ready →

Phase 2: Cutover

- Brief downtime: 2-10 min per VM
- Final incremental sync (<5% of disk)
- ECS creation: ~30 seconds
- Switch traffic to new ECS
- Decommission source after sign-off



On failure: review logs → fix issue → retry from last checkpoint

Migrate HDFS→OBS, Hive→MRS Hive, Spark→MRS Spark, Kafka→DMS Kafka+Flink. 20TB HDFS → OBS+MRS in 1 week, 50% cost reduction vs. on-prem Cloudera CDH.

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Validation & Cutover

What to Ask AI

Example: "Validate migration: compare row counts, run checksum on sample tables, execute production-equivalent workload, verify p99 latency."



AI Should Know

Data consistency checks, performance benchmarks, rollback procedures. DRS reverse sync for 24h as safety net. DAS dashboard for health metrics.



Verify & Prevent Issues

Cutover: 2-4h for 500GB, <30min for <100GB. DBA + app team joint sign-off. DNS cutover with rollback plan.



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Recommended Huawei Cloud Services

Schema & Data Tools

UGO: Schema conversion (DDL, PL/SQL)
DRS: Full + incremental data replication
Phase: Pre-migration → Migration



Target Platforms

GaussDB: OLTP (MySQL/PG compatible)
DWS: Data warehouse (MPP columnar)
MRS: Managed Hadoop/Spark/Flink



Orchestration & Storage

DataArts Studio: ELT pipeline orchestration
OBS: Bulk data transfer staging area
DAS: Database monitoring & health metrics



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Resources & GitHub Links

Database Migration Skills

Install from GitHub:
https://github.com/binrogithub/1-3-Cloud-Adoption-Skills/tree/main/AI-Assist-Migration/Database-Migration

Includes: PostgreSQL ECS→RDS via DRS, DWS Cluster Deployment



Give It Real Data

Connect AI to Huawei Cloud docs.
Provide UGO conversion reports, DRS sync status, DAS performance metrics. Share schema DDL and sample queries for context.



Set Clear Expectations

Ask for step-by-step migration plans.
Require validation at each phase: schema conversion, data sync, performance testing.
Never skip checksum validation.



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Why AI-Assisted Database Migration

Speed & Accuracy

UGO + AI converts 80% of schema automatically. DRS keeps data in sync. AI rewrites 70% of SQL. Planning that took weeks now takes days.



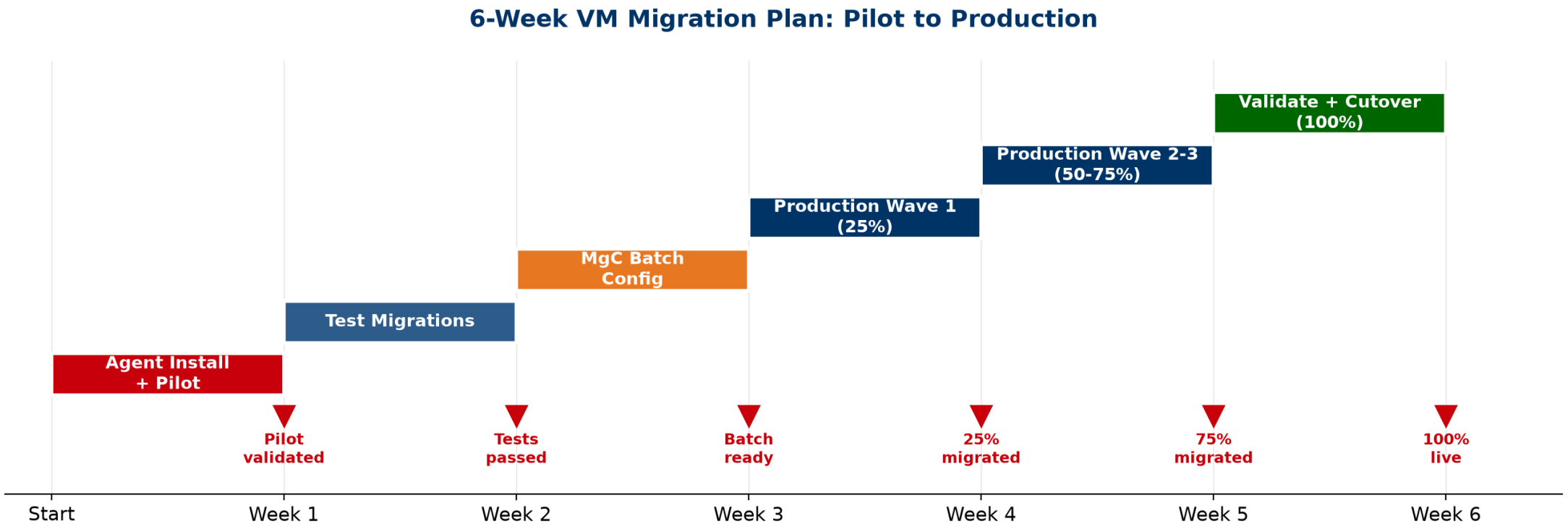
Risk Reduction

AI spots schema incompatibilities early. DRS reverse sync provides rollback safety. Checksum validation ensures 99.99% data consistency.



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Your 12-Week Database Migration Journey



Week 1: Assessment — inventory databases, classify by complexity
Week 2-3: PoC — migrate one non-critical DB end-to-end
Week 4-7: Pilot — migrate 2-3 DBs with app teams in sprints
Week 8-11: Production — batch migration with rollback plans
Week 12: Optimize — tune GaussDB/DWS, set up DR, automate backups

Source: Huawei Cloud Database Migration Best Practices

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Equip Your AI Assistant

Install Migration Skills:
https://github.com/binrogithub/1-3-Cloud-Adoption-Skills/tree/main/AI-Assist-Migration/Database-Migration

Key tools: UGO (schema), DRS (data sync), DAS (monitoring), DataArts Studio (ELT)
Contact: Huawei Cloud LATAM SMB team for migration assessment

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